

OligoTox-Kidney — Source Register

Complete provenance for every value in the dataset NIH/NCATS OligoTox Open Data Challenge, Phase 2 · Endpoint: nephrotoxicity

Generated from the dataset itself, not from recollection: every source below is cited by at least one row of data/measurements.csv or data/oligos.csv. Row and compound counts are recomputed from those files.

1. How provenance works in this dataset

Every one of the **246 measurements** carries three provenance fields, and no row exists without all three:

Field	Meaning
source_id	Short key into this register (e.g. N3, K1, A11)
source_ref	The document — DOI, patent number, PMC ID, or DailyMed SPL identifier
source_table	The exact locus inside it: table number, figure panel, or label section

A fourth field, redistribution, governs whether a raw value may be republished: public_domain **182 rows** (US patent material and US federal labels), summary_stat **64 rows** (derived figures from copyrighted publications).

Sequence provenance is tracked separately in oligos.csv via design_source and identity_confirmation, because a sequence and a toxicity value often come from different documents.

The single most important distinction in this register is §6: sources we *retrieved and read* versus sources we *cite but did not retrieve*. 36 rows rest on the latter, and they are marked in the data as well as here.

2. Databases and repositories used

Database	Base URL	Used for	Reachable from our environment
Google Patents	https://patents.google.com	Patent full text (N3, N4)	Intermittent (503s); PDFs held locally
USPTO Patent Public Search	https://ppubs.uspto.gov	Patent verification	Yes
PubMed Central (PMC)	https://pmc.ncbi.nlm.nih.gov/articles/	Open-access full text	Yes
PubMed	https://pubmed.ncbi.nlm.nih.gov/	Abstracts, PMIDs	Yes
DailyMed (NLM)	https://dailymed.nlm.nih.gov/dailymed/	US prescribing information (SPL)	Yes — our route to FDA labels
Drugs@FDA	https://www.accessdata.fda.gov/drugsatfda_docs/nda/	FDA review documents	Yes — with a browser user agent. Corrected 2026-10-03; the previous "blocked" entry was wrong. Default agent gets a 420-byte apology page; a browser agent gets the PDF. Direct document paths work even where the ...TOC.cfm index 404s
WHO INN	https://cdn.who.int/media/docs/default-source/international-nonproprietary-names-(inn)/	Chemical nomenclature → sequences	Yes

Database	Base URL	Used for	Reachable from our environment
NCBI Bookshelf	https://www.ncbi.nlm.nih.gov/books/	Monographs (NBK...)	Yes
ClinicalTrials.gov	https://clinicaltrials.gov/study/	Trial registrations (NCT...)	Yes
EMA	https://www.ema.europa.eu/en/medicines/human/EPAR/	EPARs and SmPCs	Yes
J-STAGE	https://www.jstage.jst.go.jp/	J. Toxicol. Sci.	Yes
DOI resolver	https://doi.org/	Journal articles generally	Publisher-dependent
NEJM · Circulation/AHA · ScienceDirect · Oxford Academic	—	Trial publications	NO — paywalled (403)

3. Primary sources retrieved in full — the backbone (185 rows, 75%)

N4 — US Patent 11,479,818 B2 · 81 rows · 9 compounds · public domain

"*In vitro nephrotoxicity screening assay*" (EGFR/KIM-1 readout).

<https://patents.google.com/patent/US11479818B2/en> · local copy:

[sources/US11479818_in_vitro_nephrotox_assay_patent_EGFR.pdf](#) **Locus:** Table 5, p.30 — EGFR mRNA, KIM-1 mRNA and KIM-1 protein on **rat primary PTEC**, 9 compounds × 3 concentrations × 3 biomarkers, day 3, gymnotic exposure. Values normalised to compound 1-1, **not** saline. Extracted by [scripts/extract_n4_table5.py](#), with 4 parsed cells checked against the printed table before writing.

N3 — US Patent 11,105,794 B2 · 69 rows · 21 compounds · public domain

"*In vitro nephrotoxicity screening assay*". <https://patents.google.com/patent/US11105794B2/en> · local copy: [sources/US11105794_in_vitro_nephrotox_assay_patent.pdf](#) **Two distinct loci:**

- **Table 1** (21 rows) — in-vivo nephrotoxicity grades. Method section p.25 establishes the design: "*Wistar Han Crl : WI (Han) male rats*", dosed "*at 40 mg/kg on days 1 and 8*", read on a "*Multiplex MAP Rat Kidney Toxicity Magnetic Bead Panel 2*", sacrificed "*on day 15*". This paragraph is why the species/duration/dose correction was made.
- **Table 2** (48 rows) — extracellular EGF on **primary human PTEC and PTEC-TERT1**, 3 compounds × 4 concentrations × 2 systems × 2 timepoints, % of saline.
- **SEQUENCE LISTING** — the formal listing, source of 21 sequences (validated base-for-base against the listing rather than the examples table).

M1 — Moisan et al. 2017 · 11 rows · 5 compounds · summary_stat

EGF uptake / nephrotoxicity of ASOs in vitro. PMC5363415

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5363415/> · local copy in [sources/](#) **Locus:** Table 1, Fig 1C–2, Fig 2, Fig 2C, Fig 2D — human PTEC-TERT1 intracellular ATP and extracellular EGF. **Prints no sequences** (verified) — hence AON-A/C/D/E remain sequence-TBD.

N2 — Janssen et al. 2019 · 10 rows · 3 compounds · summary_stat

Therapy with 2'-O-Me phosphorothioate ASOs causes reversible proteinuria by inhibiting renal protein

reabsorption. PMC6796739 <https://pmc.ncbi.nlm.nih.gov/articles/PMC6796739/> · local copy in [sources/](#) **Locus:** Fig 1A (study DMD114673), Fig 1D, Fig 1E, Fig 5E–G, results, 39-week necropsy. Drisapersen in DMD patients + monkey + human ciPTEC — the row set that captures the functional-not-cytotoxic phenotype.

K1 — Sandelius et al. 2020 · 9 rows · 2 compounds · summary_stat

Urinary kidney biomarker panel for ASO tubular toxicity. PMID **33084520**

<https://pubmed.ncbi.nlm.nih.gov/33084520/> · local copy in sources/ **Locus:** results, results_histopath. cEt ASO in mouse; KIM-1 / clusterin / NGAL / cystatin C. Tool and control ASO sequences are **proprietary and unpublished**.

A4 — Arch. Toxicol. 2021 · 5 rows · 1 compound · summary_stat

Nephrotoxic ASO SPC5001 induces kidney injury biomarkers in a proximal tubule-on-a-chip. DOI

10.1007/s00204-021-03062-8 · <https://doi.org/10.1007/s00204-021-03062-8> **Locus:** results — quantitative in-vitro fold-changes.

REV — Wu et al. 2022 · 4 rows · 4 compounds · summary_stat

Nephrotoxicity of marketed antisense oligonucleotide drugs. PMC**10174585**

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10174585/> · local copy in sources/ **Locus:** golodirsen / casimersen / viltolarsen / nusinersen sections. Used for cross-checking clinical grades rather than as sole primary evidence.

4. Regulatory labels and clinical anchors (A1–A13, 21 rows)

ID	Source	Locus	Rows	Link
A1	Inotersen — NEJM 2018 NEURO-TTR + FDA label 211172	label §5.2	3	https://doi.org/10.1056/NEJMOA1716793
A2	Inotersen FSGS case report — AJKD 2022	case_report	1	S0272-6386(22)00929-5
A3	SPC5001 — van Poelgeest et al. 2013	results, case_biopsy	3	https://doi.org/10.1111/bcp.12738
A5	Givosiran — NEJM 2019 ENVISION + Givlaari label	ENVISION; label	1	https://doi.org/10.1056/NEJMOA1913147
A6	Givosiran — Liver Int. 2022, ENVISION 24-month	ENVISION_24mo	1	https://doi.org/10.1111/liv.15090
A7	Nusinersen — Spinraza FDA label	label §5.3; Studies 1–2	1	DailyMed
A8	Volanesorsen — EMA SmPC Waylivra + APPROACH	SmPC §4.4/4.8	2	NCT02658175
A9	Mipomersen — FDA label 203568 + EMA EPAR Kynamro	label §5; EPAR	2	DailyMed / EMA
A10	Inclisiran — Leqvio FDA label + Novartis SmPC	label §6	1	DailyMed
A11	Golodirsen — Vyondys 53	§5.2 Kidney Toxicity	2	dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=35c227d1-5b24-44b0-b5d3-f0f6b1c46bd5

ID	Source	Locus	Rows	Link
A12	Casimersen — Amondys 45	§5.2 Kidney Toxicity	2	...setid=e9e5fd44-eeda-4580-bba1-a734828bbcc3
A13	Viltolarsen — Viltepso	§5.1 Kidney Toxicity	2	...setid=1ffff9a8-6d6a-4dcb-8493-1b6cc3a5d123

A11–A13 were read directly on DailyMed in this work. All three carry the same structure — kidney toxicity observed in animals, **not** observed in the human studies, renal monitoring nonetheless mandated (serum cystatin C, urine dipstick, UPCR). They are the dataset's best-supported human negatives.

All three also warn that **"creatinine may not be a reliable measure of kidney function in DMD patients"** because of reduced skeletal muscle mass — a caveat that applies to every DMD row.

5. Sequence provenance — WHO INN chemical nomenclature (20 sequences)

WHO publishes complete residue-by-residue nomenclature for INN-named oligonucleotides, from which sequence is recoverable by deterministic parse. Every list below was fetched as a PDF and parsed by `scripts/fill_inn_sequences.py`.

URL pattern: [https://cdn.who.int/media/docs/default-source/international-nonproprietary-names-\(inn\)/rl<NN>.pdf](https://cdn.who.int/media/docs/default-source/international-nonproprietary-names-(inn)/rl<NN>.pdf)

Recommended INN List	Compounds recovered
71	patisiran
73	mongersen, revusiran
75	fitusiran
76	cemdisiran, givosiran, inclisiran
78	teprasiran
79	alicaforsen (<i>correction entry</i>), lumasiran
80	viltolarsen
81	tofersen
83	olpasiran, vupanorsen
84	pelacarsen
85	bepirovirsen, nedosiran
86	donidalorsen, zilebesiran
88	fazirsiran

Identity confirmation across all 65 oligos: patent sequence listing 25 · WHO INN nomenclature 20 · regulatory label 7 · peer-reviewed publication 3 · not established 10.

Other sequence sources: W02016033424A1 (patent), PMC8673535 (PROMOVI), NCBI Bookshelf NBK549761 and NBK588653, and the Onpatro/Oxlumo/Amvuttra/Wainua/Qalsody labels.

6. Cited but not retrieved — 36 rows, flagged in the data

These rows were derived from **search-engine summaries** of the named documents during an earlier session whose network policy blocked full-text fetch. **The document was not opened.** They carry `source_id` = WS and

must be verified against the primary source before release.

This is disclosed rather than buried because it has a measurable consequence: among these rows, **zero reach nephrotoxicity grade ≥ 2** , against 11 of 22 anchor-sourced clinical rows (one-sided Fisher $p = 4.5 \times 10^{-11}$). Direct retrieval of 7 of them left only **one** standing as a measured negative. Full analysis in CLINICAL_VALIDATION.md.

Blocked for us — require manual acquisition:

Document	Compound	Publisher
NEJM 2015 NEJMoa1407250	mongersen	NEJM
NEJM 2023 OCEAN(a)-DOSE NEJMoa2211023	olpasiran	NEJM
NEJM 2024 NEJMoa2402478	donidalorsen	NEJM
~~Circulation 2021 CIRCULATIONAHA.120.053029~~	teprasiran	RETRIEVED 2026-10-02 via Europe PMC REST fullTextXML (PMC8487715), not via AHA
TRANSLATE-TIMI 70, Circulation 2022	vupanorsen	AHA
Yu et al. 2012, Toxicology	ISIS 113715	Elsevier
Alicaforsen review	alicaforsen	ScienceDirect
Engelhardt 2016 nat. 2015.0598 / PMID 29846725	class	Nature
KARDIA trials	zilebesiran	citation unresolvable as recorded — needs a real reference
~~FDA reviews: 211172 · 211970 · 213026 · 212154~~	inotersen, golodirsen, casimersen, viltolarsen	RETRIEVED 2026-10-03 — Pharm/Tox reviews, 22.9 MB, staged in research_staging/papers/. 217388 and 219019 not located under the probed paths; still open

Reachable — we can still retrieve these ourselves: PMC6249674 (Janas 2018) · PMC6468299 (Janas 2019) · PMC5790433 (Crooke 2018) · PMC6987735 (revusiran nonclinical) · PMC8673535 (PROMOVI) · PMC11068990 (PHYOX3) · PMC11944999 (pegaptanib) · PMC11020434 (CJASN 2024) · PMC7577764 (Clin Kidney J) · PMC12369710 (SEQUOIA) · PMC9804925 (bepirovirsen) · J-STAGE jts51_75 (3D-RPTEC) · Onpattro / Oxlumo / Rivfloza / Amvuttra / Qfitlia / Qalsody / Wainua labels via DailyMed.

7. Reference material — held but producing no rows

These inform interpretation and are **not** cited by any measurement. Held under toxicity/_shared/reference/.

- Frazier 2015, *ASO therapies review*, Toxicol. Pathol. — DOI 10.1177/0192623314551840
- Frazier 2022, *Kidney effects review*, Toxicol. Pathol. — DOI 10.1177/01926233221100414 (*in kidney sources*)
- Sioud, *Oligonucleotide immunostimulation / cytokines* (book chapter)
- Casarett & Doull, *Toxicology* (textbook)
- OligoTox Challenge brief (scope authority)
- Methods Mol. Biol.* 2022, renal-tox-in-mice chapter — NCBI NBK584232

Hepatotoxicity source panels (Dieckmann 2018 PMC5725219, Burdick 2014 NAR 10.1093/nar/gku142, Hagedorn 2013 NAT 10.1089/nat.2013.0436) are held under toxicity/hepatic/sources/ and contribute **zero rows** to this kidney dataset. Verified on every run by scripts/release_check.py.

8. How to verify any single value

- Open data/measurements.csv (or the Excel workbook) and find the row.
- Read source_id, source_ref and source_table.

- 3 Find `source_id` in §3–§4 above for the document and its link.
- 4 `source_table` gives the exact table, figure panel or label section.
- 5 For values from the two patents, the PDFs are in `sources/` — no re-acquisition needed.
- 6 For sequences, `design_source` and `identity_confirmation` in `oligos.csv` name the document and the method by which identity was established.

Every extraction is re-runnable: `scripts/extract_patent_table2.py`, `scripts/extract_n4_table5.py` and `scripts/fill_inn_sequences.py` re-parse their sources from scratch and validate against anchor values before writing.

9. Summary

	Rows	Share
Primary sources retrieved and read in full	185	75.2%
Regulatory labels read directly (incl. DailyMed)	21	8.5%
Cited but not retrieved (WS) — verify before release	36	14.6%
Review cross-checks	4	1.6%
Total	246	100%

Redistribution: `public_domain` 182 rows · `summary_stat` 64 rows. Dataset licence: **CC BY 4.0** (repository-root LICENSE). Third-party source PDFs retain their publishers' terms and are included for verification only.